Description of methods exploring added predictive signal of open-text responses above structured questionnaire items in the DAWBA depression section

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1 Data Processing

1.1 Holistic cleaning (before dev-test split)

A raw collection of the Development and Well-Being Assessment (DAWBA) dataset consisted of 348 paired adolescent-parent participant records. One duplicate record was removed. Additionally, 20 records marked as having withdrawn consent were removed, and features were subset to only include meta-data and the 84 features of the Depression section of the DAWBA, 42 pertaining to each of adolescent and parent responses. There were 3 participant meta-data features (ID, eligibility, health status) shared across each adolescent-parent pair and the 42 depression features were made up of 27 questionnaire responses (with choices {0,1} or {0,1,2}), 10 open-text responses, and 5 depression impact responses (with choices {0,1,2,3}), recorded separately for adolescent and parent responses.

1.2 Data Split + Dev only processing

The resulting data (n=327) were randomly split 67-33 into development (n=219) and held-out test (n=108) sets, stratified by eligibility group (eligible/not eligible for the study in which data was originally collected for). Adolescent and parent-reported responses were split into two individual cohorts and processed separately. All subsequent analyses are conducted separately, in parallel, on the

parent and adolescent-reported cohorts from the development set. The held-out test set was not touched after creation.

Four questionnaire prompt features (18i, 18j, 18k, and 18l) were removed due to being largely missing for both parent and adolescent cohorts. The target score for modeling was calculated as the sum of responses to the 5 depression impact questions (impact score) for each cohort. The DAWBA framework conditions prompting participants to respond to certain sections based on responses to previous sections. As such, participants not prompted to answer open-text questions were removed. Missing values for questionnaire responses in the case of not being prompted were simply marked as 0. For all other cases of missing entries in questionnaire responses (0 in adolescent and 5 in parent), the whole participant was dropped. Participants with 6 or more missing entries for text responses were dropped (17 in adolescent and 9 in parent), and answers were filled as *"Not answered"* for participants with 5 or less missing entries. This resulted in 103 records in the adolescent development cohort, and 71 in the parent. In each cohort, the sum of 23 structured questionnaire responses (S), individual structured responses (Q), and numeric representations of 10 open-text responses (T), are used to predict 1 aggregated depression impact sum score (y).

1.3 Text-derived features

Feature vectors of varying lengths were derived from open-text responses, with two main classes of derivation: previously trained logistic regression, or Large Language Model (LLM) prompting. In both classes, open-text responses were transformed into numerical representations of the presence or absence of some taxonomy of psychological domains/outcomes/symptoms.

1.3.1 Logistic Regression (LR)

A previous MLC project explored the efficacy of LLMs in annotator classification in text transcribed from interviews. In this project, a single human annotated transcripts of interviews with adolescent participants, their parents, and their therapists as part of the ImpactMe project. For each of a set of 31 outcomes, segments of text were marked as having some relevance to that outcome. The corpus of transcriptional text was then segmented in two ways. The first (*monologue*) separated participant (adolescent, parent, therapist) monologues as individual text entries, disregarding interviewer utterances. The other (*dialogue*) segmented each dialogue exchange between interviewer and participant as individual entries. In both cases, an entire segment was marked as containing some outcome if any part of it was initially marked for that outcome by the human annotator. These text segments were then embedded by Llama-3.3-70B-Instruct to generate 8192-dimensional vector representations. Logistic regression models were fit on these embeddings to predict the binary occurrence of each of 28 outcomes (3 were left out due to sparse labels), separately for each of *monologue* and *dialogue* text segmentation types, with the choice of which segmentation

type deemed a hyper-parameter for final model configuration selection. Logistic regression hyper-parameters (C regularization factor) were derived via 4-fold CV, separately for each segmentation-type/outcome pair. The set of 56 trained classifiers (one for each of 28 outcomes for each of the two segmentation types) were then available for application to data from the DAWBA.

Three methods of aggregation of the 10 open-text responses were explored: *block, max, average*. For *block*, open text-responses in the DAWBA are concatenated into blocks for each participant. For example:

"Question: Please describe your mood (sadness, irritability) and your level of interest in things? Answer: I have not felt great over the past few weeks. Question: What else has changed at the same time as your mood and level of interest? Answer: I started getting into fights with my parents...etc".

These text blocks are embedded by Llama-3.3-70B-Instruct, and passed into the trained LR models for one of the segmentation types, resulting in a 28-dimensional probability vector for each participant, serving as the numerical representation of text responses.

For *max* and *average*, each question-answer pair are embedded and passed into the LR models separately, resulting in a set of 10 28-dimensional vectors, pertaining to each of the 10 open-response questions. Note that missing responses for a given participant are marked as 0, rather than obtained from the LR model. For the *max* method, the maximum is taken over the set of 10 for each outcome, resulting in a single aggregated 28-dimensional vector. Similarly, the *average* method takes the mean over the 10 vectors for each outcome, resulting in a single 28-dimensional vector.

1.3.2 LLM Prompting

In the case of deriving numerical representations of text through LLM prompting, 3 sets of taxonomies were explored: *original, expanded, combined* . The *original* taxonomy (ImpactMe) includes the 28 outcomes explored in the LR-based method, in addition to the 3 previously left out due to sparsity, for a total of 31. The *expanded* taxonomy is a separate set of 80 outcomes, and the *combined* taxonomy is a combination of the two, totaling 111 outcomes.

Responses from the 10 open-text prompts are aggregated into blocks, similar to the *block* aggregation method for the LR-based vectors. For a given participant, this block of question-answer responses are wrapped in a broader prompt, giving context to the text and requesting the LLM to output whether or not the text contained a given outcome. Prompting was explored in two iterations.

In the first iteration, only 0-shot prompting of Llama-3.3-70B-Instruct was explored, with two separate output types: *binary, probability*. In the *binary* case, raw 0/1 output was marked for a given outcome, whereas the internal token probability of {1} was marked in the *probability* case, where output token choices were artificially limited to {0,1}. LLM temperature was set to 0.6 for all cases in this iteration.

The second iteration of prompting explored both 0 and few-shot exploration of the *expanded* taxonomy, consisting of Llama-3.3-70B-Instruct, Qwen3.6-27B, and Gemini-2.5-Pro LLMs. In addition to *binary*, a *scale* output type was explored instead of *probability*, where the LLM was prompted to rate the intensity/appearance of an outcome, with possible output set to {0,1,2,3,4}, whose values were later mapped to {0, 0.25, 0.5, 0.75, 1}. LLM temperatures of {0, 0.2, 0.4, 0.6, 0.8, 1.0} were explored for Llama-3.3-70B-Instruct and Qwen3.6-27B, whereas {0, 0.4, 0.8, 1.2, 1.6, 2.0} were explored for Gemini-2.5-Pro. An example prompt for 0-shot scale prompting is displayed in Example Prompt.

In any case, the resulting feature vector was 31, 80, and 111-dimensional for *original, expanded*, and *combined* respectively.

2 Model Development

The depression impact score (sum of depression impact responses) represents realized depression. As such, predictive performance proxies how well a set of predictors captures its indicators. Responses to the depression section of the DAWBA, including 23 structured response questions and 10 open-text response questions, were designated as linear predictors of depression impact. Although paired in the initial dataset, adolescent and parent-reported responses are processed and modeled independently.

2.1 Residual Regressor Set Up

A residual stacking framework was deployed to isolate incremental variance explained by three predictor categories. The first predictor category, denoted as $\boldsymbol{S}$, is the sum of the 23 structured responses. The second, denoted as $\boldsymbol{Q}$, is a 23-length vector of the individual structured responses to the screening questionnaire. The third, denoted as $\boldsymbol{T}$, is some numerical representation vector derived from the 10 open-text responses, aggregated based on one of the methods described in Text-derived features. In this framework, each layer of modeling is employed to predict the residuals from the previous layer of modeling, with the first layer simply predicting the target outright. The final predictions of any given layer of the stacked framework are sum of the predictions from that layer and all previous layers.

The primary stacked set-up was $\boldsymbol{SQT}$, meaning a simple linear regressor was fit on $\boldsymbol{S}$ to predict ($\hat{y_S}$) true depression impact scores (y), with residuals calculated as $\varepsilon_S = y - \hat{y_S}$. A ridge regression was then fit on $\boldsymbol{Q}$ to predict ε_S, with final $\boldsymbol{SQ}$ predictions designated as $y_{\hat{SQ}} = \hat{y_S} + \hat{\varepsilon}_S$ and residuals calculated as $\varepsilon_{SQ} = y - y_{\hat{SQ}}$. Finally, a ridge regression was fit on $\boldsymbol{T}$ to predict ε_{SQ}, with final $\boldsymbol{SQT}$ predictions designated as $y_{\hat{SQT}} = y_{\hat{SQ}} + \varepsilon_{\hat{SQ}}$. This framework is summarized in Table 1. Additional set-ups explored included single-level ($\boldsymbol{S}$, $\boldsymbol{Q}$, $\boldsymbol{T}$) and 2-layer ($\boldsymbol{ST}$, $\boldsymbol{QT}$, $\boldsymbol{SQ}$).

Table 1: Residual stacking framework by layer for SQT. ε_X denotes the residual after layer X; $\hat{\varepsilon}_X$ denotes its prediction by the next layer.

	S layer	**Q layer**	**T layer**
Fit on	Sum score S	Response vector Q	Text vector T
Predict	Impact score y	ε_S	ε_{SQ}
Incremental output	$\hat{y}_S$	$\hat{\varepsilon}_S$	$\hat{\varepsilon}_{SQ}$
Final output	$\hat{y}_S$	$\hat{y}_{SQ} = \hat{y}_S + \hat{\varepsilon}_S$	$\hat{y}_{SQT} = \hat{y}_{SQ} + \hat{\varepsilon}_{SQ}$
Residual	$\varepsilon_S = y - \hat{y}_S$	$\varepsilon_{SQ} = y - \hat{y}_{SQ}$	$\varepsilon_{SQT} = y - \hat{y}_{SQT}$

2.2 Evaluation Framework

Since this modeling framework was designed as a proxy for the ability of a set of factors to represent contributors/indicators to realized depression impact, validity of these features is represented by how well regression models are able to capture variance in depression impact. The coefficient of determination (R^2) was utilized as a primary performance metric because it measures explanatory power, that is, how much of the variance in reported depression impact is explained by a given model. This is more relevant here than others such as *MSE, RMSE,* and *MAE* because it allows for an interpretable measure of how well realized depression is captured by a model compared to a simple guess of the mean, as opposed to a raw measure of error in the model.

Because of limited dataset size, a stratified 10-repeated 3-fold cross-validation (CV) approach was utilized in the development set to explore and select final configurations. This CV framework employed a unique random seed for each of 10 repeats to split data into 3 subsets, stratified again by eligibility group. The designated metric for base evaluation of model performance was *repeat-aggregated R^2 score*, which is calculated by concatenating out-of-fold (OOF) predictions across the 3 folds within each repeat, calculating the R^2 within each repeat, then taking the average over the 10 resulting R^2 scores. This metric differs slightly from *repeat-fold-aggregated* R^2, which calculates R^2 for each set of OOF predictions, averaging over a total of 30 scores. *Repeat-aggregated* R^2 is chosen as the base metric because it is subject to lower variance, whereas *repeat-fold-aggregated* R^2 is used for formal model comparison because it allows for insight at a more granular scale.

2.3 Configurations explored

A number of modeling configurations were explored, including regression hyperparameters and text-vector derivation methods. A set of 11 lambda regularization values were explored independently in Q- and T-layer ridge regression. For example, in a SQT set up, separate values of λ were explored/selected for each of Q and T ridge regression layers. Throughout, λ subscripts denote the

model set-up in which the penalty was selected: λ_Q is the penalty for the standalone Q model, λ_{SQ} the Q-layer penalty within SQ (retained by SQT), and $\lambda_T/\lambda_{ST}/\lambda_{QT}/\lambda_{SQT}$ the T-layer penalties in T, ST, QT, and SQT respectively.

Note that Q-layer λ selections for QT and SQT did so with whatever value of λ resulted in the best performance in Q and SQ, respectively. For example, if $\lambda_{SQ} = 15$ was optimal for SQ, only configurations with $\lambda_{SQ} = 15$ were considered for SQT, although λ_{SQT} is not constrained. This ensures that signal from T supplements the best available structured-response configuration, rather than compensating for a sub-optimally regularized Q-layer. Additionally, this allows for a consistent comparison of full/reduced model pairs like SQT/SQ. In most cases, the constrained Q-layer λ is the same as the value that would have been selected without constraint.

In addition to ridge parameters, the methods of representation of open-text responses were also considered hyper-parameters of the stacked residual regressor framework. The configurations outlined in Table 2 were explored for each family of settings and are described in detail in Text-derived features. This resulted in a total of 82 total pure model configurations, each with 11 explored lambda values, for a total of 902 total whole model configurations at the T layer.

3 Results

3.1 Set Up

The two main objectives of this study are:

1. Determine if text responses carry signal in predicting depression impact incremental to what is found in structured questionnaire responses (both sum scores and granular responses).

2. Determine how incremental signal of text, in supplement of sum scores, compares to that of structured questionnaire responses

The coefficient of determination (R^2) was used as the primary metric of performance and a proxy for how well elements of the DAWBA capture depression impact. Thus, incremental R^2 serves as an indicator of the added value of any given layer of modeling.

3.2 Model Selection

Analyses in the exploratory phase serve mainly to inform confirmatory analyses that will be conducted on the held-out test set. As such, this stage does not involve formal significance testing. Rather, it employs bayesian post-hoc methods to select final model configurations and explore the viability of research questions given the train set. As such, the following exploratory analyses were conducted via 10-repeated 3-fold CV in the train set.

Table 2: Text-derivation configurations explored at the T level, by family of settings. $^{\dagger}\{0.1, 0.5, 0.75, 1, 2.5, 5, 15, 25, 50, 75, 100\}$

	ImpactMe Probs	**LLM Prompt iter. 1**	**LLM Prompt iter. 2**
LLM model	Llama-3.3-70B-Instruct	Llama-3.3-70B-Instruct	Llama-3.3-70B-Instruct, Gemini-2.5-pro, Qwen3.6-27B
Segmentation/ prompt method	Monologue/ Dialogue	0-shot	0-shot/few-shot
Temperature	NA	0.6	{0, 0.2, 0.4, 0.6, 0.8, 1.0} for Llama, Qwen; {0, 0.4, 0.8, 1.2, 1.6, 2.0} for Gemini
Aggregation/ output method	Max/Avg/Block	Binary/probability	Binary/scale
Outcome taxonomy	Original	Original/Combined	Expanded
Ridge penalty λ	11-value grid†	11-value grid†	11-value grid†
Total configs	$6 \times 11 = 66$	$4 \times 11 = 44$	$72 \times 11 = 792$

A single model configuration was selected in the train set for each of *T, ST, QT*, and *SQT*, separately for adolescent and parent cohorts. Note that a reduced model may be derived from a full model, meaning a trained SQT or ST model contains a trained and usable S model. Similarly, SQ may be derived from SQT, and Q from QT.

First, an optimal lambda value was selected for each of Q and SQ (λ_Q and λ_{SQ}, respectively) such that it produced the highest *repeat-aggregated* R^2 *score*, described in Evaluation Framework. All subsequent selections of QT and SQT configurations are constrained to those containing λ_Q and λ_{SQ} values that produced the highest *repeat-aggregated* R^2 *score* for Q and SQ, respectively.

For each selection of *T, ST, QT*, and *SQT*, configurations are ranked on *repeat-aggregated* R^2 *scores*, the top 25 of which were compared for final selection. The comparison was done using the *baycomp* package's `two_on_single()` function, which performs a bayesian correlated t-test on the 30 paired *repeat-fold-aggregated* R^2 observations, with the area of practical equivalence set to 0.01 (1% variance explained). The configuration with the highest *repeat-aggregated* R^2 *score* was compared to each of the subsequent 24 within this framework. Configurations were retained for final consideration if the comparison was either conclusive in favor of equivalence ($P(\text{equal}) > 0.5$) or inconclusive (all of $P(\text{best better})$, $P(\text{equal})$, and $P(\text{best worse})$ falling below 0.5); that is, configurations were excluded only when the comparison conclusively favored one model over the other. Probabilities were evaluated after rounding to two decimal places. Final models, displayed in Table 3, were selected from the surviving models with the following criteria, in order of importance:

1. No outlying residual error profile
    - Configurations whose residual profiles clustered apart from the main body of models were excluded, ensuring the final selection does not carry an idiosyncratic error structure relative to its peers.
2. Prompted LLM rather than ImpactMe-LR model
    - Prompting allows for more reliable equivalence between what we want to be assessed and what is actually assessed
3. LLM model preferred in following order: Qwen, Llama, Gemini
    - Qwen: open-weight, smaller than Llama
    - Llama: open-weight
    - Gemini: closed source
4. Aggregation type: few-shot, zero-shot
    - Few-shot allows for a more reliable assessment of what we want to be assessed, not as much is left to model biases
5. Lambda Regularization: higher

- Higher regularization increases reliability

6. Temperature: lower
    - More determinism increases reliability

Table 3: Final frozen model configurations selected during exploratory analysis, by model set-up and cohort. All selected configurations use the *scale* output type. Model families are Llama-3.3-70B-Instruct, Qwen3.6-27B, and Gemini-2.5-pro; *few* and *0* denote few-shot and zero-shot prompting, respectively. λ_Q and λ_T denote ridge regularization for the Q- and T-layer regressions, respectively. *Repeat-aggregated* R^2 is the development-set cross-validation performance on which selection was based.

Set-up	**Cohort**	**LLM Model**	**LLM Prompt**	$\boldsymbol{\lambda_Q}$	$\boldsymbol{\lambda_T}$	**Temp.**	**Repeat-agg. R^2**
S	Parent	–	–	–	–	–	0.182
	Adolescent	–	–	–	–	–	0.123
Q	Parent	–	–	5	–	–	0.211
	Adolescent	–	–	15	–	–	0.106
T	Parent	Llama	few	–	5	1.0	0.208
	Adolescent	Gemini	few	–	5	0.4	0.145
SQ	Parent	–	–	15	–	–	0.226
	Adolescent	–	–	50	–	–	0.141
ST	Parent	Llama	few	–	15	0.0	0.248
	Adolescent	Gemini	few	–	25	0.4	0.191
QT	Parent	Qwen	0	5	50	0.0	0.256
	Adolescent	Gemini	few	15	5	0.4	0.207
SQT	Parent	Qwen	0	15	75	0.0	0.264
	Adolescent	Gemini	few	50	15	0.4	0.211

3.3 Viability Analysis

To explore the viability of the second main objective outlined in Set Up, a similar pairwise bayesian post-hoc comparison test was employed among the final selected configurations using the *repeat-fold aggregated* R^2 results from 10-repeated 3-fold CV. Comparisons included (SQ vs ST) and (S vs Q vs T). Results prompted a one-sided confirmatory test of ST superiority over SQ, with $P(ST > SQ) = 55.7\%$ and $P(SQ > ST) = 32.7\%$ in parent responses and $P(ST$

$> SQ)$ = 79.5% and $P(SQ > ST)$ = 10.4% in adolescent responses. Further, results were not strong enough in the pairwise comparison between (S, Q, T), so this comparison will be designated down to a secondary exploratory analysis.

3.4 Outcome contribution

Given final selected model configurations, an exploratory analysis of feature contributions was performed for indication of how each feature was affecting model output. Explored in two dimensions, the salience of features and incremental R^2 provides a holistic view of each feature's effect on both raw output and aggregated performance. Because the SQT set up explains the most variance in both adolescent and parent cohorts, this was the only stacked framework in which contribution was analyzed.

3.4.1 Salience

To gain insight into how, on average, the values of features are pushing predictions up/down, given their assigned weight, a salience analysis was performed. Specifically, given a trained residual regressor model, recorded feature values were multiplied by learned weights of corresponding features. Values across records for each feature were aggregated across minimum, maximum, average, and standard deviation. These values have the ability to uncover that a feature with a relatively low/high weight magnitude is actually counterintuitively affecting model output by a large/small magnitude, given the distribution of recorded values for a given feature.

3.4.2 Average individual contribution

A salience analysis provides insight into model behavior given feature values, but an analysis of behavior given the target variable provides insight into how informative or degradative learned signal from a feature is, providing a more holistic heuristic of desirable feature contribution. The *Lindeman, Merenda, and Gold method*, also known as *LMG decomposition*, was deployed to apportion variance explained among individual predictors. The basis of this method is to iteratively add features for a model to consider and record the incremental variance explained. Because the incremental variance of a feature in this method may differ depending on what the model had already been considering, this process must be done many times with many different orderings of adding in features.

Because T enters the stacked framework as its final layer, the decomposition is applied to the T layer alone: every ordering begins from the fitted SQ prediction, so the recorded increments apportion the **incremental** variance explained by the text vector over and above sum score and structured responses, rather than the total variance explained by the SQT model. The 80 text-feature contributions therefore sum to $\Delta(R^2_{SQT}, R^2_{SQ})$ for a given LMG iteration. The S and Q layers are not apportioned. Note that this total is invariant to the

ordering in which features are added; only its allocation across features depends on the ordering. Two distinct implementations were explored.

Single-split Implementation The first, and the simpler of the two, used the same exploratory dataset that final models were trained on as an evaluation set. For a single final trained model, 500 random feature orderings were generated, each from a unique random seed, and each was evaluated alongside its reverse, for 1000 orderings in total. The process of iteratively adding features and recording evaluation-set performance was carried out for every ordering, and each text feature's incremental variance explained was averaged over all 1000 to give a single contribution value per feature. This method is heavily biased, since it is simply a measure of how much each feature helps the model memorize the training data.

CV Implementation The second implementation addressed this limitation by embedding *LMG* within the same stratified 10-repeated 3-fold CV framework described in Evaluation Framework. Here, 1000 random feature orderings were generated, each from a unique random seed, and each was again evaluated alongside its reverse, for 2000 orderings in total. A given ordering was evaluated in all 30 (repeat, fold) cells of the CV framework: within a cell, the model was fit on the two training folds and features were added iteratively, with the incremental R^2 on that cell's held-out fold recorded at each addition. Each feature's final contribution is the mean of its increment across all $1000 \times 2 \times 30 = 60{,}000$ (ordering, direction, cell) evaluations. Unlike the first implementation, contributions are estimated out of fold, so a feature whose learned signal does not generalize receives a low or negative average contribution.

3.5 Final trained model

Given configuration selections, final models trained on the entire exploratory development set will be exported for each of *T, ST, QT* and *SQT* in each of the adolescent and parent cohorts. All model configurations/hyper-parameters (text-derivation method, LLM, temperature, output type, and all lambda values) are frozen based on development-set selection before any test-set data are accessed. All confirmatory procedures retrain models using only these frozen configurations.

4 Confirmatory Analysis

Confirmatory analyses will be conducted on the held-out, and thus-far untouched, test set. As such, a single train-test split will be employed, such that models are trained once on the train set, and evaluated once on the test set. Additionally, only a single model configuration will be tested for each of *S, Q, T, SQ, ST, QT,* and *SQT*. All analyses are completed separately for parent and adolescent cohorts.

For all observed R^2 metrics in the test set, a 1000-iteration bootstrap method will be employed to generate 95% confidence intervals (CIs). In the case of a single metric (for example, the R^2 of SQT), for each of 1000 iterations, n samples from the test set will be drawn with replacement before recalculating R^2 on the sampled test set, where n is the sample size of the baseline test set. This generates a distribution of 1000 values, and the 95% CIs will be taken as the 2.5th and 97.5th percentiles of this distribution. In the case of a metric or statistic acting as a conjunction of two (for example, $\Delta(X, Y)$, where X and Y are distinct statistics), bootstrapping will be performed in a paired fashion such that records are drawn for both pieces of the conjunction, that is, the sample of participants drawn to calculate X and Y will be the same in any given bootstrap iteration. Again, this entails calculating the statistic/metric in each bootstrap iteration, and designating the 95% CI as the 2.5th and 97.5th percentile of this distribution of 1000 records. The following questions will be formally evaluated via significance testing.

4.1 Hypothesis 1A/1B

Open-text responses carry signal in predicting depression impact incremental to what is found in structured questionnaire responses. This can be represented more formally as:

$$H_A : \Delta(R^2_{full}, R^2_{reduced}) > 0$$
$$H_0 : \Delta(R^2_{full}, R^2_{reduced}) \leq 0$$

where $\Delta(x, y)$ represents the difference $x - y$, R^2_{full} is the test-set R^2 of some full model including T, and $R^2_{reduced}$ represents the test-set R^2 of some reduced model excluding T. Three full/reduced pairs will be evaluated: (ST, S), (QT, Q), and (SQT, SQ), with a single p-value derived for each pair, separately for each of the adolescent- and parent-response cohorts. In this way, *Hypothesis 1A* will test incremental test signal in the adolescent-reported cohort, whereas *Hypothesis 1B* will separately test that in the parent-reported cohort.

The null hypothesis states that T carries no predictive information, meaning that a model including T performs no better than the same model without it. To construct this null, a permutation procedure will destroy the link between each participant and their text-derived vector while leaving all other data intact. For each of 1000 iterations, a unique random seed will be used to shuffle the text-derived vectors among participants in the train set, stratified by eligibility group, such that a participant only receives the vector of another participant within the same eligibility group. Within each iteration, the same shuffled train set participants are used for all three full models: each of ST, QT, and SQT is refit on the shuffled train set and evaluated on the un-shuffled test set, yielding a null R^2 for each. The reduced models (S, Q, SQ) contain no text features and are therefore unaffected by the shuffle; their observed test-set R^2 values are held fixed across iterations, so each iteration yields three null differences $\{\Delta(R^2_{ST}, R^2_S), \Delta(R^2_{QT}, R^2_Q), \Delta(R^2_{SQT}, R^2_{SQ})\}$.

Because T should demonstrate incremental value regardless of which structured-response baseline it supplements, the three pairwise differences are treated as a family of hypotheses, with family-wise error rate (FWER) controlled at $\alpha = 0.05$ using the Westfall–Young step-down max-T procedure.

Let the observed differences be set $d = \{d_1, d_2, d_3\}$, where $d_1 = \Delta(R^2_{ST}, R^2_S)$, $d_2 = \Delta(R^2_{QT}, R^2_Q)$, and $d_3 = \Delta(R^2_{SQT}, R^2_{SQ})$, and let $d^{(i)}_j$ denote the corresponding null difference from permutation iteration i for hypothesis j. The observed differences are ordered from largest to smallest, $d_{(1)} \geq d_{(2)} \geq d_{(3)}$, where $d_{(k)}$ denotes the k-th largest observed difference in d.

Adjusted p-values are computed step-down. The null statistic for step $k = 1$ in permutation iteration i is $m^{(i)}_1 = \max_j d^{(i)}_j$: the maximum value across all three permuted differences within iteration i. The corresponding hypothesis of $d_{(1)}$ is then removed from consideration for further testing. Accordingly, for step $k = 2$ in permutation iteration i, $m^{(i)}_2$ is the maximum across the two remaining hypotheses ($\{d^i_{(2)}, d^i_{(3)}\}$), and $m^{(i)}_3$ is simply the single remaining permuted difference ($d^i_{(3)}$).

The raw adjusted p-value for the k-th ordered hypothesis is

$$\tilde{p}_{(k)} = \frac{1}{1000 + 1} \left(1 + \sum_{i=1}^{1000} \mathbf{1}\{m^{(i)}_k \geq d_{(k)}\} \right),$$

after which monotonicity is enforced by taking successive maxima across all p-values, $p_{(k)} = \max\{\tilde{p}_{(1)}, \ldots, \tilde{p}_{(k)}\}$. Because all three null differences within an iteration are computed from a single shared shuffle, the permutation distribution preserves the dependence among the hypotheses, so the procedure adapts to their correlation rather than assuming independence, serving to increase power over conservative alternative FWER controls such as the Bonferroni correction. Significance is set at the standard $\alpha = 0.05$, and Hypothesis 1 will be supported in a given cohort if at least one of the three adjusted p-values meets the $p < 0.05$ threshold.

1000-iteration paired bootstrapping, described in Confirmatory Analysis, will be used to generate 95% CIs for each of $\{\Delta(R^2_{ST}, R^2_S), \Delta(R^2_{QT}, R^2_Q), \Delta(R^2_{SQT}, R^2_{SQ})\}$ in each cohort, regardless of significance.

4.2 Hypothesis 2A/2B

Does the incremental signal of open-text responses, in supplement of sum scores, exceed that of granular structured questionnaire responses?

This question can be represented more formally as:

$$H_A : \Delta(R^2_{ST}, R^2_{SQ}) > 0$$
$$H_0 : \Delta(R^2_{ST}, R^2_{SQ}) \leq 0$$

where $\Delta(x, y)$ represents the difference $x - y$, and R^2_{ST} and R^2_{SQ} represent the test-set R^2 of the stacked ST and SQ regressor models, respectively. Similar

to **Hypothesis 1A/1B**, *Hypothesis 2A* will refer to a separate test of the adolescent cohort, whereas *Hypothesis 2B* will do the same in the parent cohort, separately.

A one-sided test is pre-specified based on the direction indicated by Bayesian post-hoc comparisons in the development set during exploratory analyses. Unlike **Hypothesis 1A/1B**, the null hypothesis here does not assert that T carries no information. Under H_0, both T and Q may carry signal, and the claim is only that ST does not outperform SQ, meaning a train-set shuffle would therefore construct the wrong null. Instead, a per-participant prediction-swap (sign-flip) procedure will be employed, under which the null of exchangeability states that the assignment of the labels *"ST"* and *"SQ"* to each participant's pair of predictions is arbitrary.

The frozen ST and SQ configurations will each be trained once on the full development set, producing a pair of test-set predictions $(\hat{y_{ST,i}}, \hat{y_{SQ,i}})$ for each of the n test participants. The observed statistic is computed on these baseline predictions as $t_{obs} = \Delta(R^2_{ST}, R^2_{SQ})$. For each of 1000 iterations, a unique random seed will be used to swap $\hat{y_{ST,i}}$ and $\hat{y_{SQ,i}}$ independently for each participant i with probability 0.5. Both R^2 values are then recomputed on the swapped prediction vectors and their difference recorded as t_i. Note that models are trained only once, as permutation applies exclusively to test-set predictions. This is in contrast to the **Hypothesis 1A/1B** procedure, which retrains models under each train-set shuffle. Significance will once again be set at $\alpha = 0.05$, and p-value defined as:

$$p = \frac{1}{1000 + 1} \left(1 + \sum_{i=1}^{1000} 1\{t_i \geq t_{obs}\} \right).$$

A single p-value for each of 2A and 2B will be calculated, with significance once again set at the standard $p < 0.05$. 1000-iteration bootstrapping will also be utilized to generate a 95% CI of $\Delta(R^2_{ST}, R^2_{SQ})$ for each cohort, regardless of significance.

4.3 Note on multiple tests

Hypothesis 1 and 2 each have a pair of A/B tests associated with them, each evaluated separately in adolescent- and parent-report cohorts. Although not statistically independent, since there exists overlap in paired responses, each addresses a distinct respondent perspective and constitutes a separate claim. All four results will be reported in full, and no conclusion will be extended from one of the four tests to another. Accordingly, no correction will be applied across cohorts (A/B) or hypothesis classes (Hypothesis 1/2).

5 Further Exploratory Analyses

Secondary exploratory analyses will also be performed in addition to confirmatory analyses. These analyses will not be used to make any conclusions about

the data, rather, potentially form new confirmatory hypotheses.

One exploration will derive feature-impact through a framework similar to that described in **Salience**. Weights from frozen models trained on development data will be multiplied by recorded values in the held-out test set and aggregated. These values will be compared to those in the development set.

Another exploration will employ an LMG method similar to that described in **Single-split Implementation**, where frozen models trained on the development set will be applied to the test-set over 1000 iterations (2000, including reversals) of feature-orderings to obtain average contribution values for each of the 80 psychological features. These values will be compared to the development-set results described in **CV Implementation**.

Additionally, exploratory sensitivity analyses will be performed for insight into any potential biases or unforeseen relationships between test-set performance and aspects of the data. For example, we will look at the pairwise relationship between performance and sex, missingness, and/or date of questionnaire completion. An analysis on the characteristics of excluded participant records will also be performed to uncover potential biases caused by our filtering framework. Further, a comparison of single-layer S, Q, and T model test-set performance will be performed, as there was no clear trend in predictive ability across adolescent and parent cohorts in the development set.

6 Other

6.1 Limitations

This analysis aims to explore if structured questionnaire responses are insufficient in completely capturing contributors to realized depression impact, and if so, if missing information may be extracted from open-text responses. This may seem to create somewhat of a paradox, since reported depression impact is taken from the same source/modality we may be claiming is insufficient. We argue that, since participants are asked about their realized experiences outright, as opposed to a screening for auxiliary emotions and experiences, depression impact questions operate in a separate framework from those in the screening questionnaire. If that is not convincing enough to accept impact sum score as a sufficient representation of realized depression, it may be more digestible to understand it as a representation of at least some slice of realized depression. In this case, it is still relevant to effectively capture contributors to what the DAWBA is screening for.

Additionally, the sample size is relatively small, increasing variance and limiting power.

6.2 Example Prompt

The following is an example 0-shot scale prompt for the *irritability* outcome of a parent-reported open text response. Only two question/answer pairs are

included here, but actual prompts contain all 10, even if some (up to 2) are left "Not answered".

```
### Task

Your task is to analyze the parent's responses to a questionnaire (at
most 10 questions) and rate the severity of irritability associated
with his/her child, by answering the question below.


The input contains responses from a parent, structured as question/
answer pairs (up to Q10/A10). Each answer is written by the parent and
 may contain relevant behavioral observations about the child.


---

### Instructions

- Consider both explicit statements and reasonable implicit
indications.

- Base your judgment only on the information provided in the parent's
responses.

- Evaluate the responses either collectively or each question
independently, as appropriate.

- Use reasonable clinical inference when evidence is indirect or
implied.

- Respond with exactly one token from the provided options.

- Do not generate any additional text.

- The output must exactly match one of the provided response options.

---

Definition of Irritability: A persistent tendency to react to stimuli
with feelings of frustration, annoyance, or anger, often
disproportionate to the situation. It can manifest as a low tolerance
for frustration, heightened sensitivity to perceived provocations, and
 a tendency toward outbursts (verbal or behavioral) that are hard to
control. Irritability may present continuously (a generally short fuse
 or grumpy baseline mood) or episodically (discrete outbursts
triggered by specific events), and can be directed at others (snapping
, arguing, aggression) or turned inward (tension, restlessness,
frustration).
```

```
Question: Does the child show signs of increased irritability, a short
 temper, or anger outbursts (e.g., snapping at others, low frustration
 tolerance, or difficulty calming down once upset)?

---

### Question:

#### Select the most appropriate response

- 0 - Absent: the parent's responses contain no implicit or explicit
indication of irritability associated with the child.

- 1 - Very low: the parent's responses contain a faint, indirect, or
highly ambiguous indication of irritability associated with the child.

- 2 - Low to moderate: the parent's responses contain a clear but
subdued indication of irritability associated with the child, present
but not salient.

- 3 - Moderate to high: the parent's responses contain a strong and
clearly expressed indication of irritability associated with the child
, salient but not dominant.

- 4 - Very high: the parent's responses contain an explicit, dominant,
 or pervasive indication of irritability associated with the child.

Note: Rate the severity of the symptom, not the number of mentions.

Implicit cues (tone, wording, context) should be considered.

---

### Questionnaire:

Q1: How would you describe your child's mood over the past two weeks?

A1: He's been snapping at his siblings a lot more than usual.

Q2: Has your child mentioned feeling overwhelmed or anxious?

A2: Not directly, but he seems more withdrawn after school.

---

### Response:
```